DIGITAL FORENSICS INVESTIGATION REPORT

Memory Forensics — Windows 10 Endpoint Compromise

Case Name	Reveal Lab — Windows Memory Forensics Investigation
Case Number	2026-010
Examiner	Tej Chaute
Date	07/04/2026
Platform	CyberDefenders
Evidence File	192-Reveal.dmp (Memory Dump)— SHA256: 7e724ff06c0967416958752aee8569c4bbc3ea733d572a9d870ebbfedcdf553d
Classification	Training

1. Case Summary

A memory forensics investigation was conducted on a Windows 10 (Build 19041, 64-bit) system following a SIEM alert flagging suspicious activity. Examination of the memory dump using Volatility 3 identified a multi-stage, fileless attack leveraging legitimate Windows utilities for execution and evasion.

The attack originated from a transient parent process (PID 4120) that spawned both a decoy application (wordpad.exe) and a malicious PowerShell instance. PowerShell executed hidden commands to access a remote WebDAV share hosted on 45.9.74.32, from which a second-stage payload (3435.dll) was executed using rundll32.exe. Network analysis confirmed active communication with attacker-controlled infrastructure over TCP port 8888. Memory analysis revealed injected executable regions with RWX permissions, indicating reflective DLL loading and in-memory execution. The observed tactics, techniques, and procedures align with the StrelaStealer malware family, known for stealthy, fileless credential-stealing operations.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	192-Reveal.dmp	7e724ff06c0967416958752aee8569c4bbc3ea733d572a9d870ebbfedcdf553d	2.00 GB

3. Tools Used

Tool	Version	Purpose
Volatility 3	2.28.1	Primary memory analysis — process listing, network connections, command-line extraction, injection detection
strings	2.44	Raw ASCII/Unicode string extraction from memory dump
grep	3.11	Filtering and pattern matching against extracted strings

4. Key Findings

Each finding is supported by at least one artifact. Source tool, Volatility plugin, and PID context are documented per item.

4.1 System Information

Field	Value
OS	Windows 10 (Build 19041.1, amd64)
Kernel Version	10.0.19041
Architecture	x64
Number of Processors	2
System Time (UTC)	2024-07-15 07:00:08 UTC
Evidence Type	Full Memory Dump
Analysis Tool	Volatility 3

4.2 Artifact Table

#	Artifact Type	Location / Volatility Plugin	PID / Sequence	TimeStamp (UTC)	What It Proves
1	Malicious Process	windows.pslist	3692	2024-07-15 07:00:03	Identifies the presence of powershell.exe executing at the time of compromise. PowerShell is commonly abused by attackers for script-based execution, indicating potential malicious activity rather than normal user behavior.
2	Process Relationship	windows.psscan	3692 / 4120	2024-07-15 07:00:03	Reveals that powershell.exe and wordpad.exe were spawned by the same parent process (PID 4120), which is no longer present. This indicates a transient launcher or malicious script initiating multiple processes simultaneously, a behavior consistent with staged attacks.
3	Command Execution	windows.cmdline	3692	2024-07-15 07:00:03	Shows that PowerShell was executed with -windowstyle hidden, confirming attempts to evade user visibility. The command includes net use and rundll32, demonstrating automated execution of additional payloads without user interaction.
4	Remote Share Access	Cmdline + strings	3692	2024-07-15 07:00:03	Confirms access to a remote WebDAV share (\\45.9.74.32\davwwwroot). This indicates that the attacker used external infrastructure to host and deliver malicious payloads instead of storing them locally.
5	Payload Execution	windows.pstree	3692	2024-07-15 07:00:03	Demonstrates execution of 3435.dll using rundll32.exe, confirming the presence of a second-stage payload. This reflects the use of legitimate Windows binaries (LOLBins) to execute malicious code.

#	Artifact Type	Location / Volatility Plugin	PID / Sequence	TimeStamp (UTC)	What It Proves
6	Network Connection	windows.netscan	2416	2024-07-15 07:00:06	Identifies an active outbound connection from the compromised host to 45.9.74.32:8888 via net.exe. This confirms communication with attacker-controlled infrastructure and supports the presence of command-and-control or payload delivery activity.
7	User Context	windows.getsids	3692	2024-07-15 07:00:03	Shows that the malicious process was executed under user **Elon**, who has administrative privileges. This indicates that the attacker operated within a legitimate user context, increasing the potential impact of the compromise.
8	Memory Injection	windows.malware.malfind	3692	N/A	Reveals memory regions with PAGE_EXECUTE_READWRITE permissions, indicating injected or dynamically generated code. This confirms in-memory execution and reflective DLL loading, commonly used to evade disk-based detection mechanisms.

4.3 Volatility Commands Run

All commands below were run against 192-Reveal.dmp.

```
python3 vol.py -f 192-Reveal.dmp windows.info
python3 vol.py -f 192-Reveal.dmp windows.pslist
python3 vol.py -f 192-Reveal.dmp windows.pstree
python3 vol.py -f 192-Reveal.dmp windows.psscan
python3 vol.py -f 192-Reveal.dmp windows.cmdline
python3 vol.py -f 192-Reveal.dmp windows.getsids --pid 3692
python3 vol.py -f 192-Reveal.dmp windows.malware.malfind --pid 3692
python3 vol.py -f 192-Reveal.dmp windows.netscan
strings 192-Reveal.dmp | grep 45.9.74.32
```

5. Event Timeline

Reconstructed chronological sequence of events based on Volatility artifact timestamps. All times in UTC. Entries marked '~' indicate events inferred from process relationships where precise timestamps were not independently recoverable.

Timestamp (UTC)	Artifact Source	Event Description
2024-07-15 07:00:03 ~	windows.psscan (Volatility 3)	A transient and unidentified parent process (PID 4120) executed and spawned multiple child processes. The absence of this process in active listings indicates it was short-lived, suggesting a dropper or script-based launcher used to initiate the attack.
2024-07-15 07:00:03	windows.pslist (Volatility 3)	The malicious process powershell.exe (PID 3692) was launched. This marks the beginning of attacker-controlled execution on the system, likely triggered by the transient parent process.
2024-07-15 07:00:03	windows.cmdline (Volatility 3)	PowerShell executed a hidden command using the -windowstyle hidden flag, indicating deliberate evasion of user visibility and interactive detection.
2024-07-15 07:00:03	windows.cmdline + strings	The system accessed a remote WebDAV share (\\45.9.74.32\davwwwroot) using the net use command, establishing a connection to attacker-controlled infrastructure for payload staging.
2024-07-15 07:00:03	windows.pstree (Volatility 3)	The second-stage payload (3435.dll) was executed via rundll32.exe, confirming the use of a legitimate Windows utility to load and execute malicious code from a remote source.
2024-07-15 07:00:06	windows.netscan (Volatility 3)	A network connection was established from the host to 45.9.74.32:8888 using net.exe, confirming active communication with the attacker's server and reinforcing the presence of command-and-control or payload delivery activity.
2024-07-15 07:00:08	windows.info (Volatility 3)	System time recorded via windows.info, used to validate and anchor the timeline of events observed during the investigation.

6. Conclusion

The investigation confirms the system was compromised through a multi-stage, fileless attack leveraging legitimate Windows utilities throughout the execution chain.

Initial Access and Execution: A short-lived parent process (PID 4120) spawned both wordpad.exe as a decoy and powershell.exe (PID 3692) as the primary attack vector. The simultaneous spawning of a legitimate-looking application alongside the malicious process is consistent with distraction or masquerading techniques used in staged attacks. PowerShell was invoked in hidden window mode to suppress user visibility.

Payload Delivery: The PowerShell command used net use to mount a remote WebDAV share at \\45.9.74.32\davwwwroot and then invoked rundll32.exe to execute 3435.dll directly from the remote share. This approach — using two LOLBins in sequence — avoided writing any executable to disk, significantly reducing the forensic footprint and evading file-based detection.

Command and Control: An active outbound TCP connection to 45.9.74.32:8888 was identified under net.exe (PID 2416). The association of a raw TCP connection on a non-standard port with net.exe is anomalous and may indicate process name spoofing or unusual LOLBin repurposing for C2 communication.

Defense Evasion: malfind identified RWX memory regions containing shellcode patterns within the PowerShell process, confirming reflective DLL loading and in-memory execution. No corresponding payload file exists on disk, making host-based detection dependent on memory forensics rather than file scanning.

Threat Attribution: The combination of WebDAV-based delivery, PowerShell hidden execution, rundll32 LOLBin abuse, and in-memory reflective loading aligns with the StrelaStealer malware family, known for stealthy, fileless credential-stealing operations targeting email clients.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path and represents the first Windows memory forensics case in this portfolio using Volatility 3.

Key methodological observations from this case:

- The transient parent process (PID 4120) was not recoverable from active process listings but appeared in psscan output — psscan scans raw memory for EPROCESS structures and can find processes that

have exited or been unlinked from the active list. Running both pslist and psscan is essential; relying on pslist alone would have missed the parent relationship entirely.

- The simultaneous spawning of wordpad.exe alongside powershell.exe from the same parent is a recognised decoy technique. Analysts who see a benign-looking application in a process tree may stop investigating that branch — the decoy is designed to reduce scrutiny on the malicious sibling process.
- net.exe establishing an outbound TCP connection to port 8888 is inherently suspicious. net.exe is a Windows network utility designed for SMB, share management, and user/group administration — it has no legitimate reason to initiate raw TCP connections to external IPs on non-standard ports. This process-to-connection pairing warrants escalation regardless of whether the destination IP is known-malicious.
- Reflective DLL loading leaves no file on disk — the only evidence is in memory. PAGE_EXECUTE_READWRITE regions in a process that should not be self-modifying are the key indicator. malfind is effective at surfacing these but produces false positives; the RWX regions need to be assessed in context of the process and whether they contain recognisable shellcode patterns.
- WebDAV shares are a common payload delivery mechanism precisely because Windows natively supports them and outbound HTTP/HTTPS traffic to external shares typically passes through corporate firewalls without inspection. Monitoring net use commands targeting external IPs is a high-signal detection opportunity.

If this were a real-world investigation, recommended next steps would include: isolating the host immediately given active C2 communication; recovering 3435.dll from the WebDAV server or memory dump for static and dynamic analysis; identifying what credentials StrelaStealer may have harvested; reviewing email client data for exfiltrated content; and hunting for lateral movement from the compromised Elon account given its administrative privileges.